

Thermal Bridge Calculation Report **Project:** Window Reveal Renovation (Fensterlaibung)

Date: 2025-12-18

Method: 2D Finite Difference Method (FDM) according to ISO 10211

Executive Summary

The thermal analysis compares the existing building condition (“Altbau”) with two renovation scenarios for the window reveal: 1. **Standard Renovation:** External Wall Insulation (WDVS) + New Window, but **NO** insulation on the reveal (Masonry Rebate only). 2. **Optimized Renovation:** External Wall Insulation (WDVS) + New Window + **30mm Reveal Insulation**.

Assessment

- **Case 3 (Optimized with 30mm Reveal Insulation)** is the recommended solution. It raises the critical internal surface temperature to **17.1°C**, eliminating condensation and mold risk ($f_{Rsi} = 0.724 > 0.70$).
 - **Case 2 (Standard Renovation without Reveal Insulation)** is borderline safe. The surface temperature reaches **13.2°C** ($f_{Rsi} = 0.698$), which is just below the recommended safety limit of 0.70, leaving a residual risk of mold growth if indoor humidity is high.
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Detailed Results

Simulation Parameters

- **Interior Temperature:** 20.0 °C
- **Exterior Temperature:** -5.0 °C
- **Wall Construction:** solid Brick ($\lambda = 0.81$).
- **Window System:** $U_f = 1.3 \text{ W/m}^2\text{K}$, $U_g = 1.1 \text{ W/m}^2\text{K}$.
- **Geometry:** High-Detail Frame including Masonry Rebate (Fensteranschlag) and Sash (Flügel).

Results Table

Scenario	Description	Ψ Value [W/mK]	f_{Rsi} Factor	Min Temp (Wall)	Min Temp (Frame)	Min Temp (Glass)
Case 1	Istzustand (36cm)	-0.121	0.574	9.4 °C	10.2 °C	13.1 °C

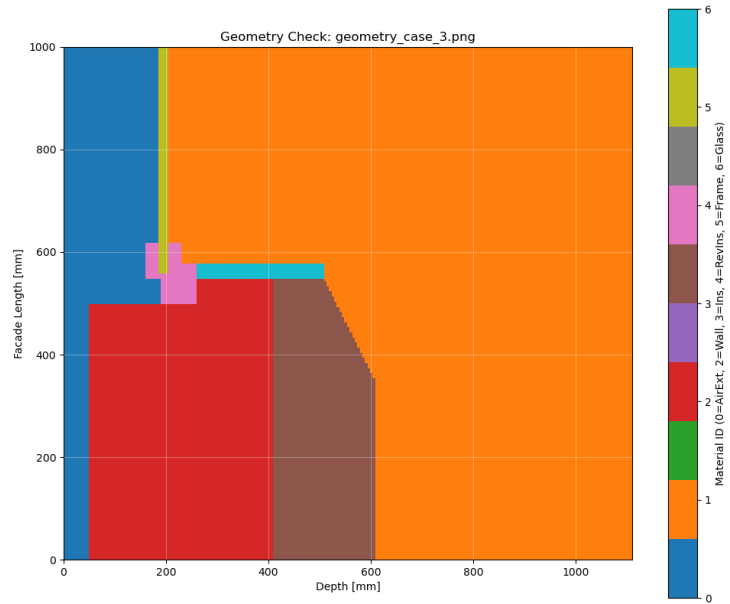
Scenario	Description	Ψ Value [W/mK]	f_{Rsi} Factor	Min Temp (Wall)	Min Temp (Frame)	Min Temp (Glass)
Case 2	Dämmung+0.174 Fas- sade (No Rev Ins)		0.698	13.2 °C	12.5 °C	13.1 °C
Case 3	Dämmung-0.017 Fas- sade und Lai- bung (30mm Rev Ins)		0.724	17.1 °C	14.6 °C	13.1 °C

Note on Ψ -Values: Negative Ψ -values indicate that the actual heat loss of the corner is *less* than the simplified 1D surface calculation would suggest (Geometry Bonus).

Visual Documentation

Case 3: Optimized Geometry (30mm Reveal Insulation)

The insulation (purple) continuously wraps the corner, keeping the masonry



(red/orange) warm.

Temperature Field (Isotherms)

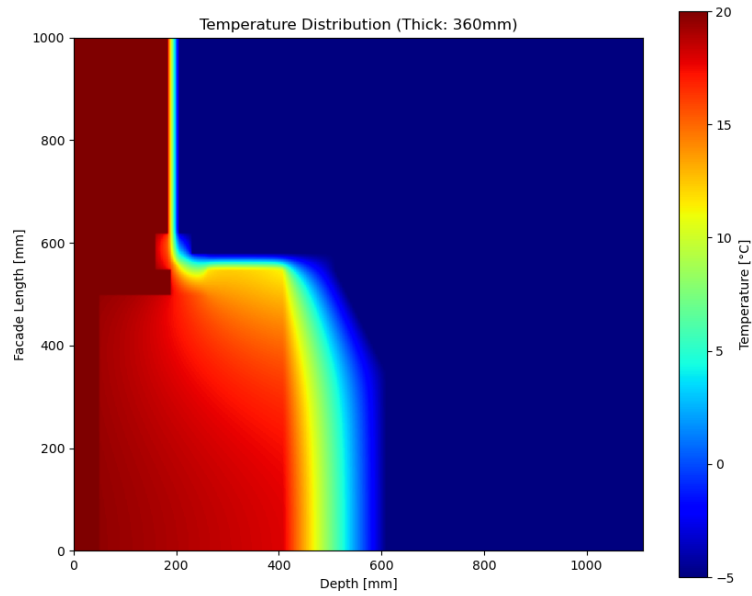
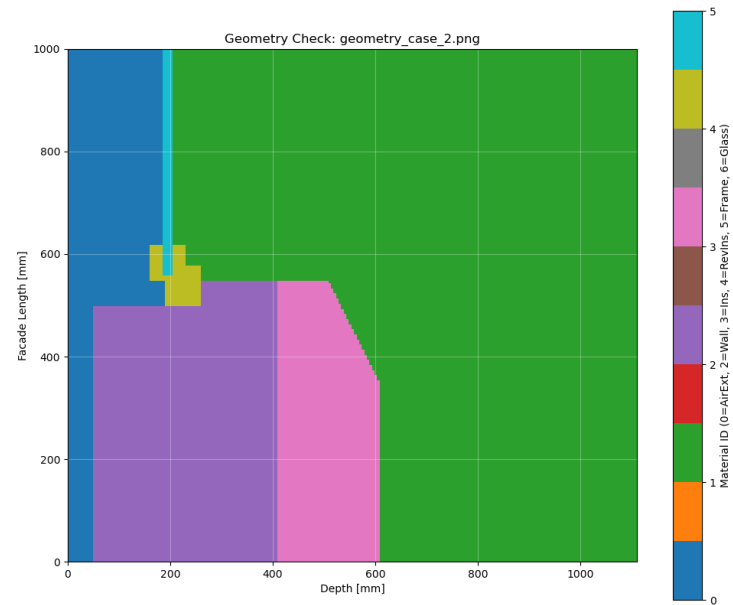


Figure 1: Case 3 Temperature

Case 2: Standard Geometry (No Reveal Insulation)



The masonry rebate remains uninsulated.

Conclusion

To ensure long-term freedom from mold and condensation, the execution of **Case 3 / Case 6 (30mm Reveal Insulation)** is strongly advised. The standard execution without reveal insulation (Case 2) does not provide a robust safety margin.